

Gagan Bhatia

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in [gaganbhatiaml](https://www.linkedin.com/in/gaganbhatiaml)

🔗 [gagan3012](https://github.com/gagan3012)

[Google Scholar](https://scholar.google.com/citations?user=9333333333333333)

EXPERIENCE

QATAR COMPUTING RESEARCH INSTITUTE | RESEARCH SCIENTIST

Doha, QA | Sept 2025 – March 2026

- Designed agentic RAG architectures with structured tool-calling for iterative evidence retrieval and answer revision, achieving SOTA faithfulness.
- Contributed retrieval and verification components to a multi-agent QA platform deployed in production, serving 1.9M+ requests in under a year.

UNIVERSITY OF ABERDEEN | RESEARCH ASSISTANT

Aberdeen, UK | Mar 2025 – Oct 2025

- Implemented attention-based interpretability methods to deliver actionable insights for improving LLM architecture. (**Date Fragments**)
- Designed Agentic AI frameworks and workflows for efficient data extraction from large documents in medical and financial domains

UNIVERSITY OF BRITISH COLUMBIA | RESEARCH ASSISTANT

Vancouver, CA | Sept 2022 – Sept 2024

- Engineered and trained a foundation model (**Qalam**) for Arabic OCR, achieving a 3-point F1 score improvement over SOTA and creating a deployable asset for digital humanities.
- Built and benchmarked multimodal LLMs (**FinTral**) using distributed training on a multi-GPU cluster.

XTRACT ONE | MACHINE LEARNING ENGINEER

Toronto, CA | Oct 2021 – Sept 2022

- Owned the full lifecycle of a Vision Transformer (ViT) system for image manipulation detection, from training to deployment via a REST API on AWS for a national defense client.
- Architected and deployed a scalable document intelligence pipeline on AWS processing over 10,000 documents daily, automating OCR, table, and text extraction.

AMAZON WEB SERVICES | SOFTWARE ENGINEER

Vancouver, CA | May 2021 – Dec 2021

- Enhanced the core NLU model for AWS Lex, improving intent recognition accuracy by 10% through targeted fine-tuning, directly impacting product performance for end-users.
- Deployed a production-ready XGBoost model on AWS SageMaker for a healthcare application, automating the detection of critical symptoms from streaming data.

EDUCATION

UNIVERSITY OF TECHNOLOGY NUREMBERG | PH.D. NLP

Nuremberg, DE | 2026 – 2029

UNIVERSITY OF ABERDEEN | MSc. ARTIFICIAL INTELLIGENCE

Aberdeen, UK | 2024 – 2025

- Thesis:** Distributional Semantics Tracing: A Framework for Explaining Hallucinations in Large Language Models

UNIVERSITY OF BRITISH COLUMBIA | BSc. ECE

Vancouver, CA | 2018 - 2022

SKILLS

Languages

PYTHON, C++, JAVA, SQL, Go, SWIFT

AI/ML Frameworks

PYTORCH, TENSORFLOW, JAX, HUGGING FACE, SCIKIT-LEARN, LANGCHAIN

AI/ML Concepts

LLM FINE-TUNING (LoRA, QLoRA), RLHF, NLP, COMPUTER VISION, MLOPS

Tools & Platforms

AWS (SAGEMAKER, S3, EC2), GCP, DOCKER, KUBERNETES, GIT, CI/CD, VECTOR DBs

SELECTED PUBLICATIONS

DATE FRAGMENTS: A HIDDEN BOTTLENECK OF TOKENIZATION FOR TEMPORAL REASONING [↗](#)

EMNLP 2025

Gagan Bhatia, Maxime Peyrard, Wei Zhao

FROM RAG TO AGENTIC RAG FOR FAITHFUL ISLAMIC QUESTION ANSWERING [↗](#)

ACL 2026

Gagan Bhatia, Hamdy Mubarak, Mustafa Jarrar, George Mikros, Fadi Zaraket, et al.

SWAN: DIALECT-AWARE, ARABIC-CENTRIC CROSS-CULTURAL EMBEDDING MODELS AND BENCHMARKS [↗](#)

NAACL 2025

Gagan Bhatia, El Moatez Billah Nagoudi, Abdellah El Mekki, Fakhraddin Alwajih, Muhammad Abdul-Mageed

FINTRAL: POWERFUL MULTIMODAL FINANCIAL LARGE LANGUAGE MODELS [↗](#)

ACL 2024

Gagan Bhatia, El Moatez Billah Nagoudi, Hasan Cavusoglu, Muhammad Abdul-Mageed